

MUHAMMAD USMAN ZAFAR

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[Usman Zafar | LinkedIn](#) | [Github Profile](#) | [Work Portfolio](#)

PROFESSIONAL SUMMARY

Computer Science graduate specializing in Machine Learning and Artificial Intelligence, with hands-on experience building AI-powered mobile applications, backend services, and data-driven models. Skilled in Python, Kotlin, and REST API development, with a strong foundation in computer vision, NLP, and reinforcement learning. Seeking an entry-level role as an AI/ML Engineer, Android Developer, Backend Developer, or Software Engineer.

SKILLS

Programming Languages: C++, Python, Java, Kotlin

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Scipy, Matplotlib, Keras, OpenCV, TF-IDF, Reinforcement Learning, NLP, Computer Vision.

Mobile & Backend Development: Android Studio, Kotlin, Jetpack Compose, FastAPI, MongoDB, REST APIs.

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Android Studio, Antigravity.

Core Competencies: Problem Solving, Team Collaboration, Technical Documentation, Continuous Learning.

PROJECTS

INFITRON – AI-Based Smart Wardrobe & Outfit Recommendation System | *Final Year Project*

Python, Kotlin, FastAPI, MongoDB, TensorFlow, PyTorch, MediaPipe, FashionCLIP

- Developed an AI-powered smart wardrobe application featuring clothing digitization, automated classification, smart tagging, wardrobe management, and AI virtual try-on.
- Built a reinforcement learning-based outfit recommendation engine with weather, calendar, and user preference integration, along with AR visualization and voice assistant support.
- Engineered a FastAPI backend with MongoDB and implemented wardrobe analytics to deliver a scalable, data-driven user experience.

Heart Disease Classification & Model Comparison | *Python, Scikit-learn, Pandas*

- Developed a ML pipeline using PCA for dimensionality reduction and compared Logistic Regression, KNN, SVM, Random Forest, and ANN models using Accuracy, Precision, Recall, F1-score, and ROC-AUC to identify the best-performing classifier.

Toxic Comment Classification System | *Python, TF-IDF, Random Forest, Gradio*

- Developed an NLP-based toxic comment classification system using TF-IDF and Random Forest, deployed with Gradio for real-time predictions and evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

House Price Prediction System | *Python, Scikit-learn*

- Developed a regression model with data preprocessing and feature engineering, evaluated using standard regression metrics.

Truck Dispatching Mobile App | *Android Studio, Java*

- Built a logistics management Android application for delivery assignment and real-time tracking with an intuitive user interface.

EDUCATION

BS Computer Science, University of Central Punjab, Lahore

Oct 2022 – July 2026

Major: Machine Learning & Artificial Intelligence

CERTIFICATIONS

- AWS Academy Cloud Foundations – Amazon Web Services (AWS)